

# CO2xNxPhrag: Porewater Sulfide

September 2026 Samples

2026-09-28

## Code Set up

### run information

```
##Before or After Reruns (YES or NO)
After_Reruns = "NO"
#If NO will not write final data file
#And Will Write a Rerun File

###things that need to be changed

sample_month = "September"
sample_year = "2026"
Run_Date = as.Date("09/17/2026", format = "%m/%d/%Y")
Run_by = "Melanie Giessner" #Instrument user
Script_run_by = "Melanie" #Code user

Run_notes="" #any notes from run

#Import Plate Data

folder_path_rawdata <- paste0("Raw Data")
Raw_plates <- list.files(path = folder_path_rawdata, full.names = TRUE, pattern = ("(202609|2026010)(.+Plate)"))
print(Raw_plates)

## [1] "Raw Data/20260917_LTREB_H2S_Plate1.csv"
## [2] "Raw Data/20260917_LTREB_H2S_Plate2.csv"
## [3] "Raw Data/20260917_LTREB_H2S_Plate3.csv"
## [4] "Raw Data/20260917_LTREB_H2S_Plate4.csv"
## [5] "Raw Data/20260917_LTREB_H2S_Plate5.csv"
## [6] "Raw Data/20260918_LTREB_H2S_Plate6.csv"
## [7] "Raw Data/20260918_LTREB_H2S_Plate7.csv"
## [8] "Raw Data/20260918_LTREB_H2S_Plate8.csv"

# Define the file path for QAQC log file - NO Need to change just check year
log_path <- here("Porewater QAQC Logs", "GCRew_H2S_QAQC_Log.csv")

# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCRew_CO2xNxPhrag_Porewater_H2S_202609.csv"
rerun_path <- "Processed Data/GCRew_CO2xNxPhrag_Porewater_H2S_202609_reruns.csv"

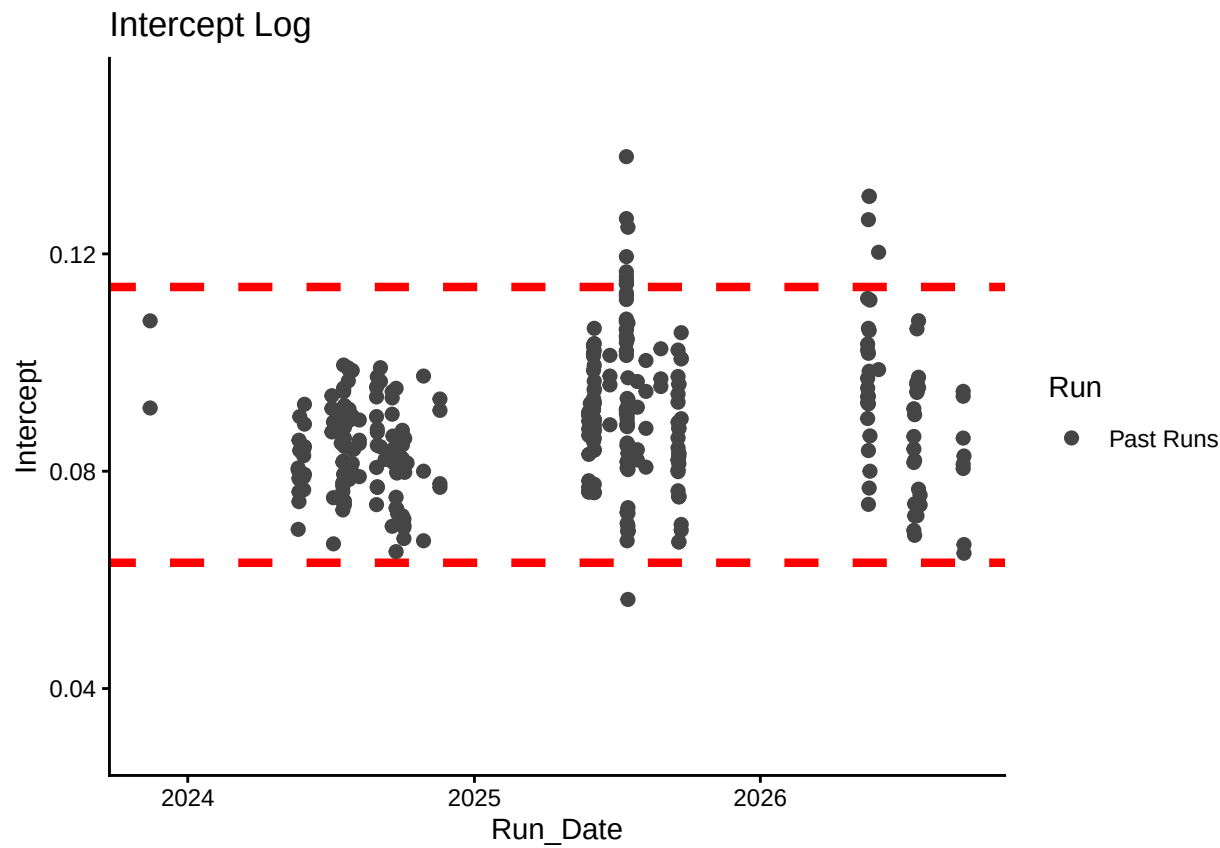
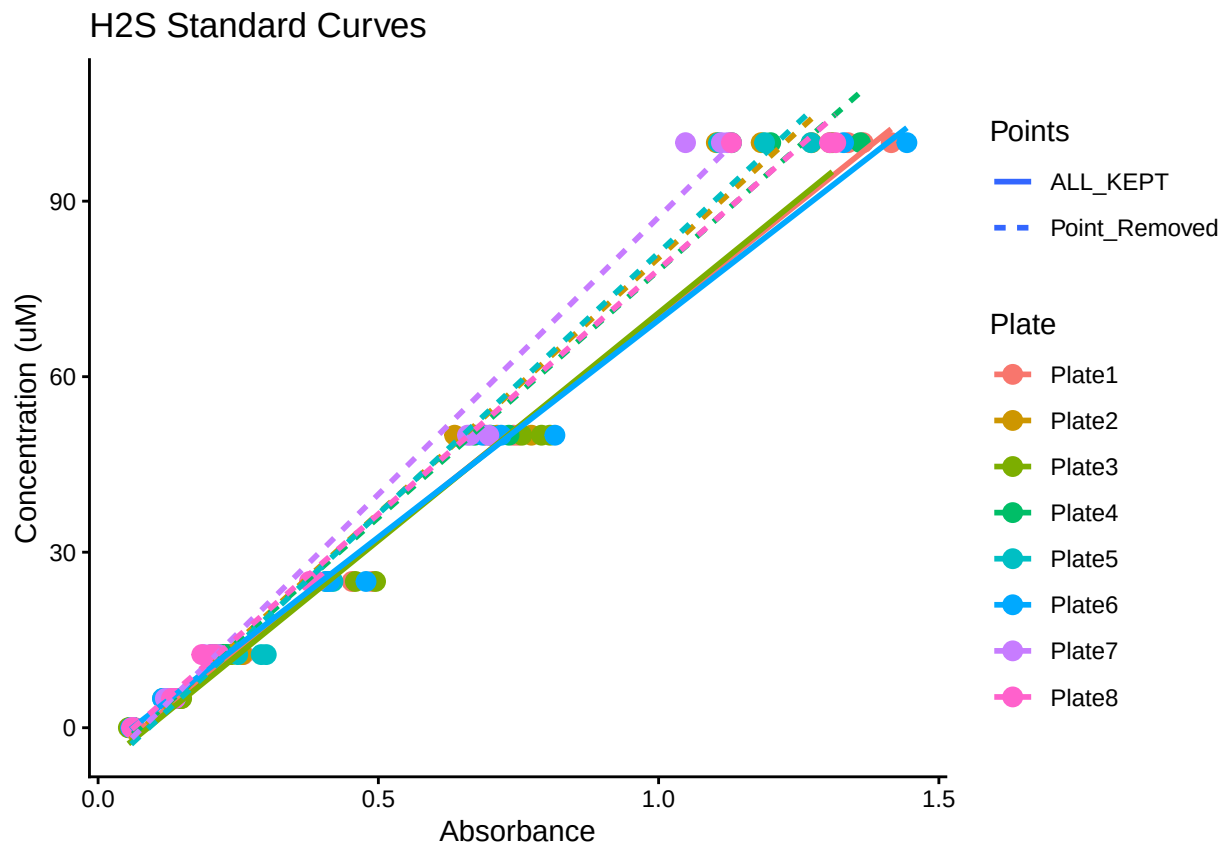
##Read in metadata and create similar sample IDs for matching to samples
##Read in data
```

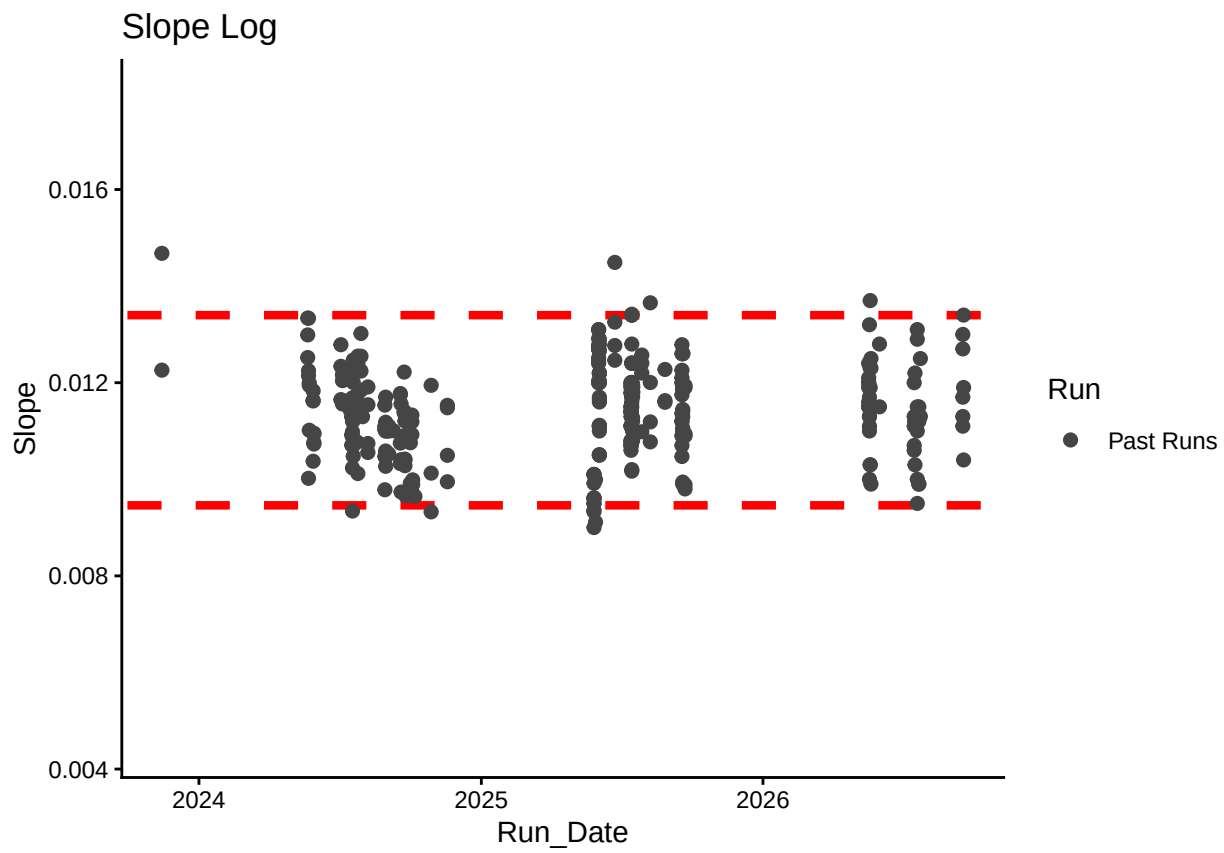
```
##Remove High CV Stds
```

```
##Check R2 & Remove Standards if Necessary
```

```
## Joining with 'by = join_by(Plate, IDs)'
```

Plot Standard Curves



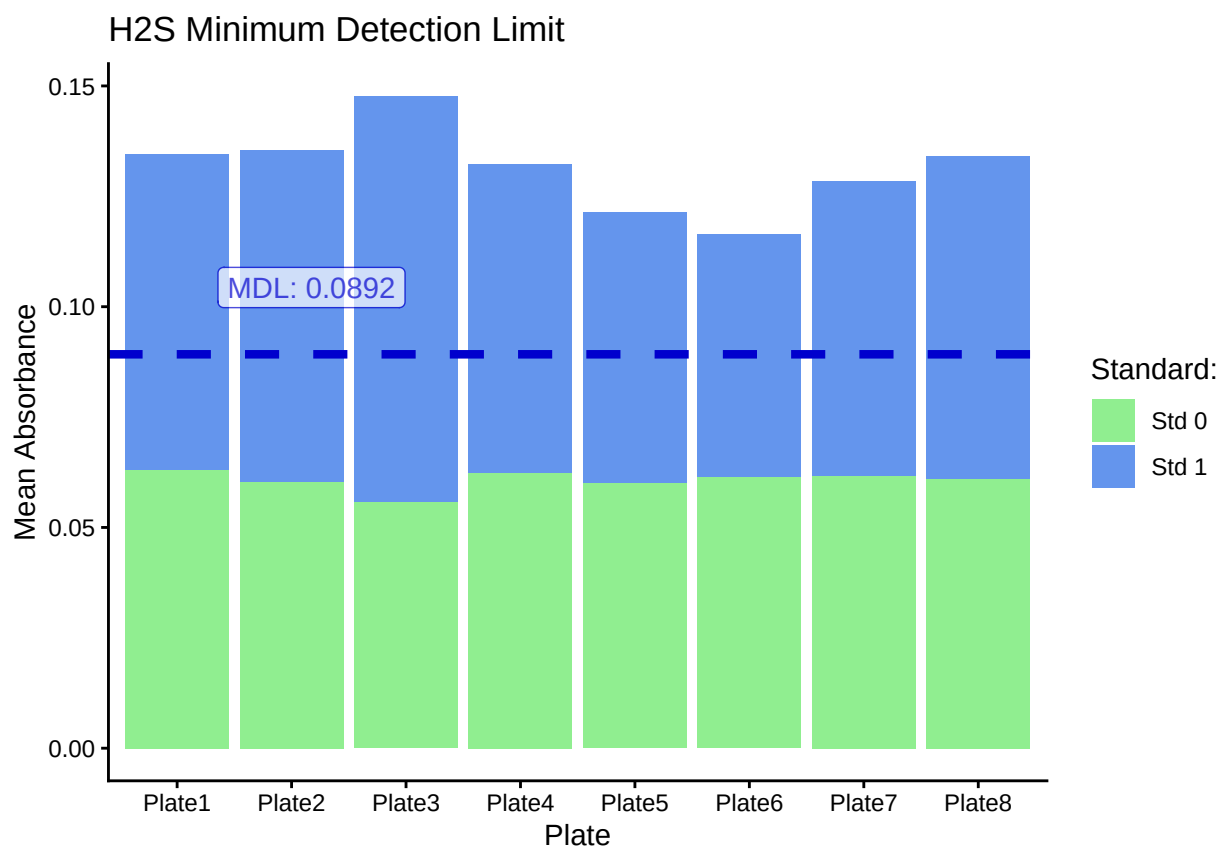


### Return Flags For Standard Curve

| Plate  | R2_Flag | CV_Flag |
|--------|---------|---------|
| Plate1 | R2_Good | CV_Good |
| Plate2 | R2_Bad  | CV_Good |
| Plate2 | R2_Bad  | CV_High |
| Plate3 | R2_Good | CV_Good |
| Plate4 | R2_Good | CV_Good |
| Plate5 | R2_Good | CV_Good |
| Plate6 | R2_Good | CV_Good |
| Plate7 | R2_Good | CV_Good |
| Plate8 | R2_Good | CV_Good |

| Plate  | Curve_RMVD |
|--------|------------|
| Plate1 | KEPT       |
| Plate2 | REMOVED    |
| Plate3 | KEPT       |
| Plate4 | KEPT       |
| Plate5 | KEPT       |
| Plate6 | REMOVED    |
| Plate7 | KEPT       |
| Plate8 | KEPT       |

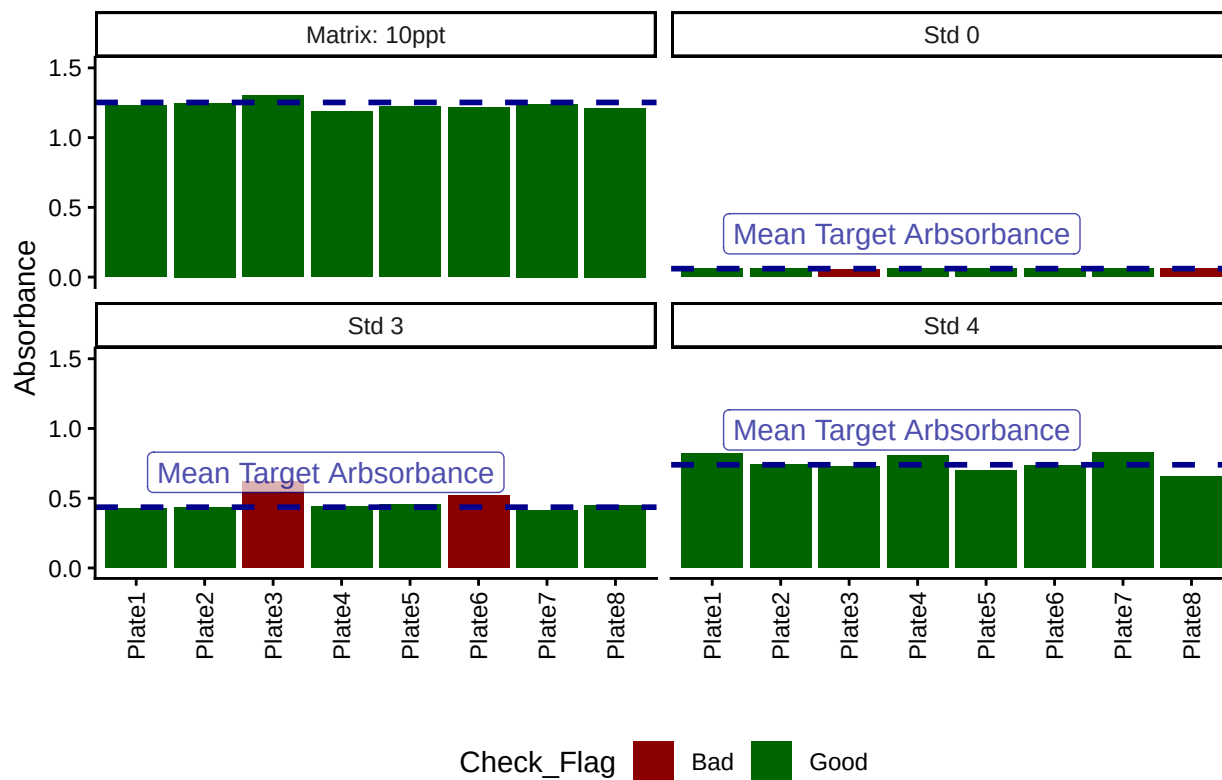
## Method Detection Limit



## Compare Check Standards to Standards

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_label_repel()').
```

## Check Stds: H2S with Confidence Interval



## Display Any Check Flags

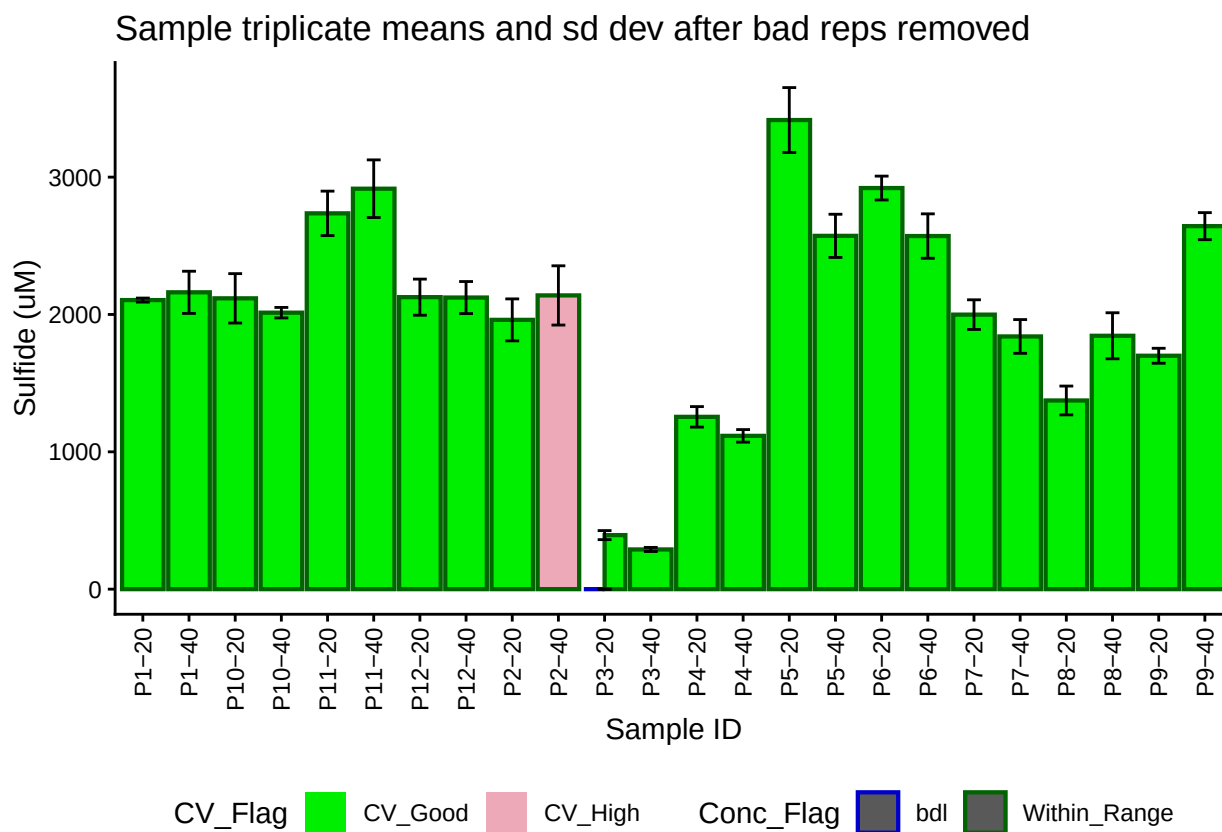
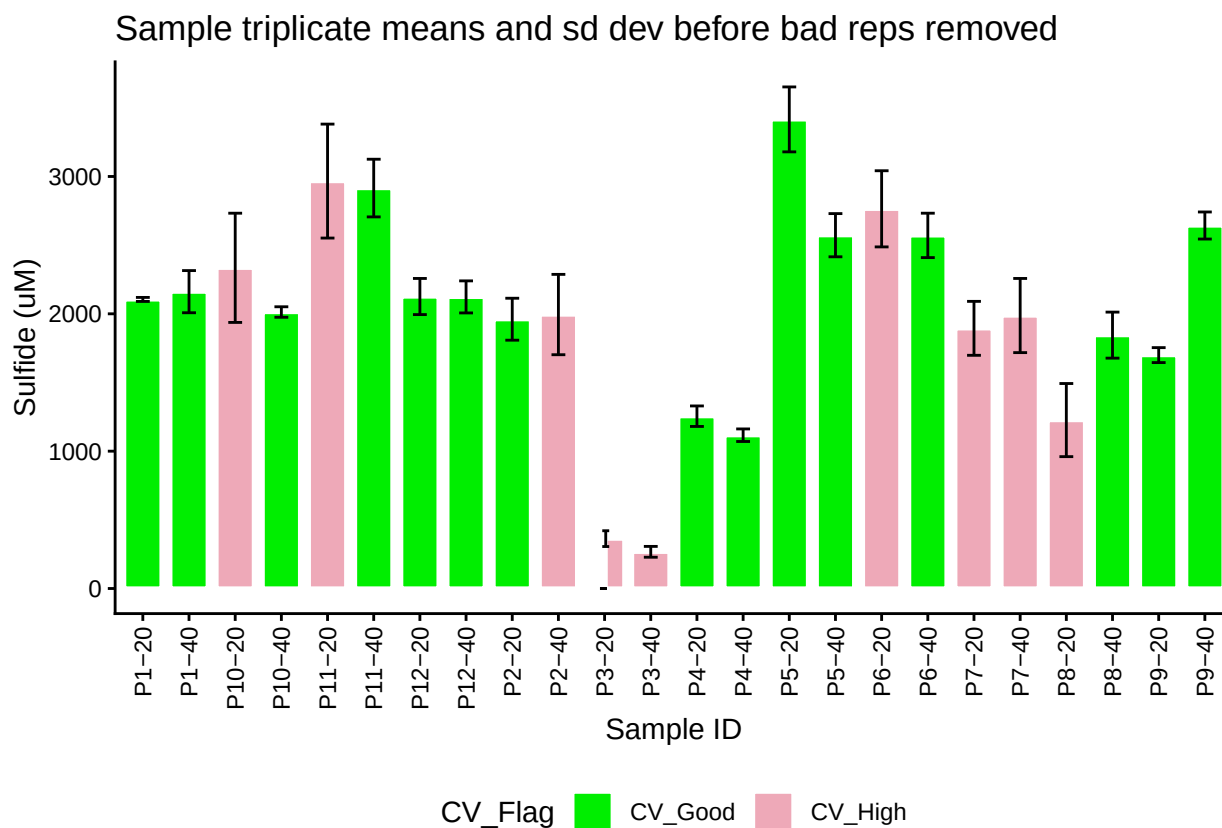
| Plate  | CHKs_Flag   |
|--------|-------------|
| Plate3 | Std0Chk_Bad |
| Plate3 | Std3Chk_Bad |
| Plate6 | Std3Chk_Bad |
| Plate8 | Std0Chk_Bad |

| Plate  | CHK_CV_Flags    |
|--------|-----------------|
| Plate8 | Std3Chk_CV_High |

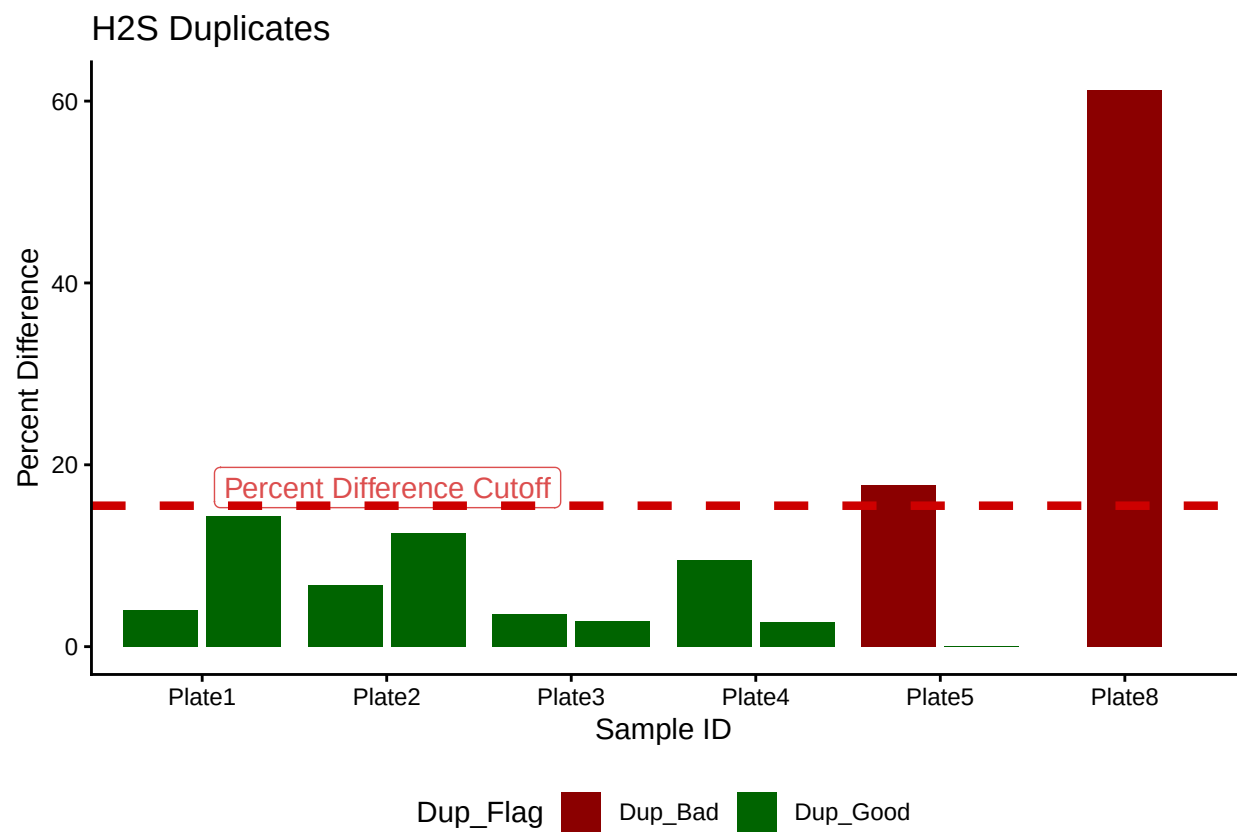
##Calculate Sulfide Concentrations & Add Concentration Flags

##Remove High CV Samples and Average

## Plot Samples



Duplicate QAQC

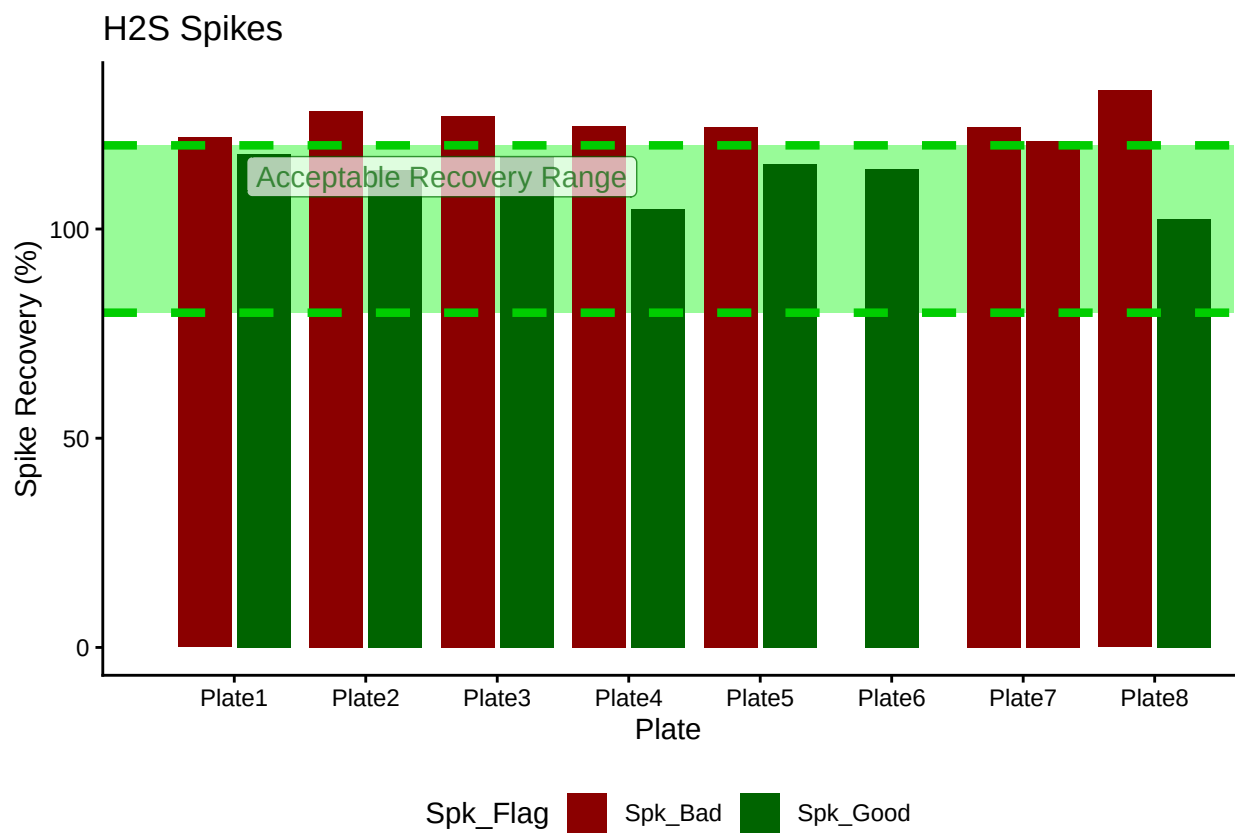


Display Dup Flags

## [1] ">60% Dups Pass"

| Plate  | Dup_Flag |
|--------|----------|
| Plate1 | Dup_Good |
| Plate2 | Dup_Good |
| Plate3 | Dup_Good |
| Plate4 | Dup_Good |
| Plate5 | Dup_Bad  |
| Plate8 | Dup_Bad  |

Spike QAQC



Display Spike Flags

```
## [1] "Spike Missing Pair"
```

| Plate  | Spike_Flag |
|--------|------------|
| Plate1 | Spk_Bad    |
| Plate2 | Spk_Bad    |
| Plate3 | Spk_Bad    |
| Plate4 | Spk_Bad    |
| Plate5 | Spk_Bad    |
| Plate6 | Spk_Good   |
| Plate7 | Spk_Bad    |
| Plate8 | Spk_Bad    |

```
## [1] "<60% of Spikes Pass, Rerun"
```

Merge samples with Metadata & check all samples are present

```
## All sample IDs are present in data
```

```
##Add Flags
```

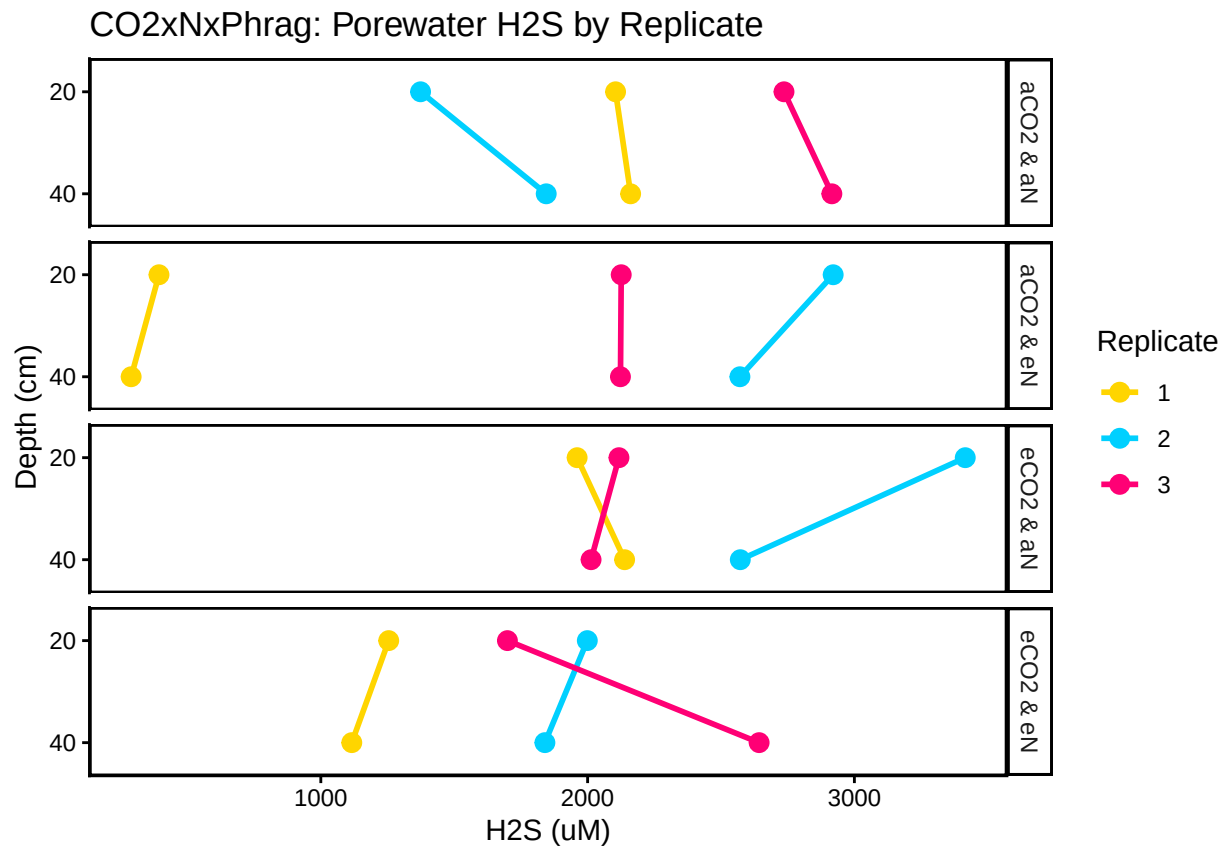
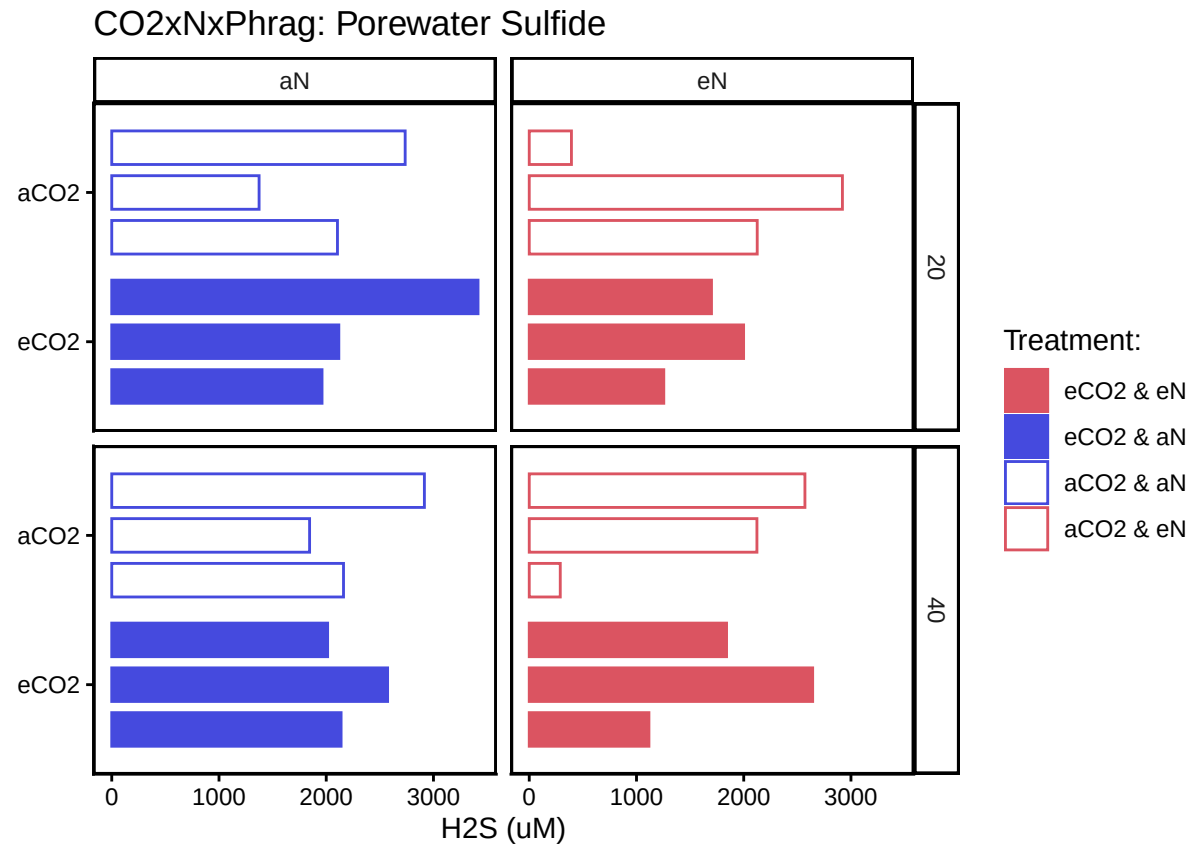
```
##Organize Data
```

```
## Adding missing grouping variables: 'Sample_ID.x'
```

```
## Adding missing grouping variables: 'Sample_ID.x'
```

Visualize Data

## Visualize Data



##Export Data

Table 7: Reruns

| Sample_ID.x | Conc     | Conc_Flag    | CV_Flag | Dilution | CV       |
|-------------|----------|--------------|---------|----------|----------|
| P2-40       | 2138.701 | Within_Range | CV_High | 50       | 10.08685 |

END